Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double Cross #2

Physical address of dairy:

Number and Street	City	County	Zip Code
18637 E Bradbury RD	Turlock	Merced	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1984

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X910-X001-XXXX X045-X910-X004-XXXX

B. OPERATORS

Ahlem, Mark

Operator name: Ahlem, Mark

Telephone no.: (209) 668-0807 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Ahlem, Mark

Legal owner name: Ahlem, Mark

Telephone no.: (209) 668-0807 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95324

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	53	0	857	0	0
Number under roof	658	0	11	0	0	0
Maximum number	658	53	11	857	0	0
Average number	658	53	11	857	0	0
Avg live weight (lbs)	1,150	1,200	650	500		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 24,787.09 *tons per reporting period*

Total nitrogen from manure: 306,667.76 *lbs per reporting period*

After ammonia losses (30% loss applied): 214,667.43 *lbs per reporting period*

Total phosphorus from manure: 51,462.76 *lbs per reporting period*

Total potassium from manure: 123,419.24 *lbs per reporting period*

Total salt from manure: 322,006.65 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 27,363,000 *gallons*

Total nitrogen generated: 55,883.93 *lbs*

Total phosphorus generated: 16,483.66 *lbs*

Total potassium generated: 110,773.69 *lbs*

Total salt generated: 667,848.68 *lbs*

	36,159,000	*gallons applied*
+	0	*gallons exported*
-	8,796,000	*gallons imported*
=	27,363,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation Well	Ground water
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/01/2023	Corral solids *SM-GLR*	270.00 *ton*	Dry-weight	41.4	21,900.00	100.00	14,300.00		45.10
01/01/2023	Corral solids *SM-GLR*	200.00 *ton*	Dry-weight	41.4	21,900.00	100.00	14,300.00		45.10

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
02/21/2023	Process wastewater *WW-GLR*	432,000.00 *gal*	566.00	127.00	1,760.00		7,700
04/19/2023	Process wastewater *WW-GLR*	360,000.00 *gal*	148.00	52.90	484.00		3,120
04/21/2023	Process wastewater *WW-GLR*	1,320,000.00 *gal*	148.00	52.90	484.00		3,120
08/05/2023	Process wastewater *WW-GLR*	1,980,000.00 *gal*	207.00	65.90	257.00		1,600
08/14/2023	Process wastewater *WW-GLR*	2,400,000.00 *gal*	207.00	65.90	257.00		1,600
09/12/2023	Process wastewater *WW-GLR*	2,304,000.00 *gal*	207.00	65.90	257.00		1,600

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/14/2023	Solid commercial fertilizer *NH3*	3.60 *ton*	Dry-weight	0.1	82.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	5,898.10	0.00	0.00	0.00
Dry manure	12,063.40	55.08	7,877.01	248,428.84
Process wastewater	15,661.40	4,875.25	27,465.30	160,744.73
Total imports for all materials	33,622.89	4,930.33	35,342.31	409,173.57

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Airport	17	17	2	manure	X044-X023-X017-XXXX X044-X026-X012-XXXX X044-X026-X014-XXXX
DC #1	17	17	2	process wastewater	X045-X091-X001-XXXX
DC #2	43	43	2	process wastewater	X045-X091-X004-XXXX
DC #4	78	78	2	process wastewater	X045-X091-X017-XXXX X045-X091-X018-XXXX
Freeway	30	30	2	manure	X044-X032-X009-XXXX X044-X032-X010-XXXX X044-X032-X011-XXXX
Fullumer	20	20	1	manure	X044-X050-X029-XXXX
Greenway	43	43	2	manure	X044-X026-X014-XXXX X044-X027-X007-XXXX
Lavender	19	19	1	manure	X041-X020-X005-XXXX X044-X051-X003-XXXX
Tucker	27	27	1	manure	X044-X051-X002-XXXX
Totals for areas that were used for application	294	294	15		
Totals for areas that were not used for application					
Land application area totals	294	294	15		

B. CROPS AND HARVESTS

Airport

Field name: Airport

Airport

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	278.00 *ton*	As-is		57.6	5,000.00	1,200.00	11,000.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.35	163.53	39.25	359.76	1,303.53

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	288.00 *ton*	As-is		61.9	4,900.00	600.00	5,400.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	16.94	166.02	20.33	182.96	1,290.92

DC #1

Field name: DC #1

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	339.00 *ton*	As-is		66.6	5,200.00	1,100.00	8,900.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.94	207.39	43.87	354.95	1,078.98

DC #1

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	509.31 *ton*	As-is		64.4	3,900.00	500.00	5,400.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.96	233.68	29.96	323.56	2,090.45

DC #2

Field name: DC #2

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	900.00 *ton*	As-is		66.6	5,200.00	1,100.00	8,900.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.93	217.67	46.05	372.56	1,132.49

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,155.52 *ton*	As-is		60.3	4,500.00	700.00	5,600.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.87	241.85	37.62	300.97	1,941.65

DC #4

Field name: DC #4

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 78 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,352.00 *ton*	As-is		66.6	5,200.00	1,100.00	8,900.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.33	180.27	38.13	308.53	937.87

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 78 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,954.20 *ton*	As-is		57.1	4,500.00	600.00	6,000.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.05	225.48	30.06	300.65	1,977.65

Freeway

Field name: Freeway

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	497.00 *ton*	As-is		57.6	5,000.00	1,200.00	11,000.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.57	165.67	39.76	364.47	1,320.56

Freeway

06/15/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 30 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	224.55 *ton*	As-is		43.0	12,800.00	1,200.00	6,100.00		14.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.49	191.62	17.96	91.32	1,271.40

Fullumer

Field name: Fullumer

01/01/2023: Alfalfa Haylage

Crop: Alfalfa Haylage Acres planted: 20 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	269.09 *ton*	As-is		50.1	14,700.00	1,200.00	3,100.00		17.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	22.50	450.00	40.50	315.00	0.00
Total actual harvest content	13.45	395.56	32.29	83.42	2,336.40

Greenway

Field name: Greenway

Greenway

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	640.00 *ton*	As-is		57.6	5,000.00	1,200.00	11,000.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.88	148.84	35.72	327.44	1,186.41

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	751.21 *ton*	As-is		61.9	4,900.00	600.00	5,400.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	17.47	171.21	20.96	188.68	1,331.21

Lavender

Field name: Lavender

01/01/2023: Alfalfa Haylage

Crop: Alfalfa Haylage Acres planted: 19 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	367.00 *ton*	As-is		60.4	10,200.00	1,000.00	4,800.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	22.50	450.00	40.50	315.00	0.00
Total actual harvest content	19.32	394.04	38.63	185.43	1,743.98

Tucker

Field name: Tucker

01/01/2023: Alfalfa Haylage

Crop: Alfalfa Haylage Acres planted: 27 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	279.63 *ton*	As-is		51.1	14,200.00	1,000.00	3,800.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	22.50	450.00	40.50	315.00	0.00
Total actual harvest content	10.36	294.13	20.71	78.71	840.69

NUTRIENT BUDGET

A. LAND APPLICATIONS

Airport - 10/27/2022: Oats, silage-soft dough

Field name: Airport

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	189.60	17.70	49.30	939.15	272.00 *ton*
Application event totals		189.60	17.70	49.30	939.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	307.78	2,985,642.00 *gal*
Application event totals		0.00	0.00	0.00	307.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	243.16	2,358,835.00 *gal*
Application event totals		0.00	0.00	0.00	243.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	213.58	2,071,875.00 *gal*
Application event totals		0.00	0.00	0.00	213.58	

Airport - 06/08/2023: Corn, silage

Field name: Airport

Crop: Corn, silage — Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Airport - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	161.65	24.71	35.01	902.96	442.00 *ton*
Application event totals		161.65	24.71	35.01	902.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	230.38	2,234,812.00 *gal*
Application event totals		0.00	0.00	0.00	230.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	208.49	2,022,454.00 *gal*
Application event totals		0.00	0.00	0.00	208.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	32.80	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	199.04	1,930,822.00 *gal*
Application event totals		32.80	0.00	0.00	199.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	32.80	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	177.60	1,722,892.00 *gal*
Application event totals		32.80	0.00	0.00	177.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	195.12	1,892,760.00 *gal*
Application event totals		0.00	0.00	0.00	195.12	

Airport - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	206.94	2,007,416.00 *gal*
Application event totals		0.00	0.00	0.00	206.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	165.63	1,606,721.00 *gal*
Application event totals		0.00	0.00	0.00	165.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	182.94	1,774,621.00 *gal*
Application event totals		0.00	0.00	0.00	182.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	163.43	1,585,421.00 *gal*
Application event totals		0.00	0.00	0.00	163.43	

DC #1 - 10/26/2022: Oats, silage-soft dough

Field name: DC #1

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.24	7.81	56.99	618.51	180,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	330.19	3,203,055.00 *gal*
Application event totals		48.24	7.81	56.99	948.70	

DC #1 - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.60	6.25	45.59	494.81	144,000.00 *gal*
Application event totals		38.60	6.25	45.59	494.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.53	13.04	186.61	774.02	216,000.00 *gal*
Application event totals		58.53	13.04	186.61	774.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.27	9.42	134.78	559.02	156,000.00 *gal*
Application event totals		42.27	9.42	134.78	559.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	26.15	9.35	85.53	551.36	360,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	219.16	2,126,003.00 *gal*
Application event totals		26.15	9.35	85.53	770.52	

DC #1 - 06/15/2023: Corn, silage

Field name: DC #1

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

DC #1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.59	15.08	138.72	789.34	600,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	273.78	2,655,813.00 *gal*
Application event totals		43.59	15.08	138.72	1,063.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	235.28	2,282,425.00 *gal*
Application event totals		0.00	0.00	0.00	235.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.43	20.97	80.72	507.18	630,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	207.25	2,010,425.00 *gal*
Application event totals		54.43	20.97	80.72	714.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.09	33.55	129.15	811.49	1,008,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	198.62	1,926,774.00 *gal*
Application event totals		87.09	33.55	129.15	1,010.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.16	28.96	111.46	700.39	870,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	163.62	1,587,211.00 *gal*
Application event totals		75.16	28.96	111.46	864.01	

DC #1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.03	8.49	32.67	205.29	255,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	192.99	1,872,140.00 *gal*
Application event totals		22.03	8.49	32.67	398.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.88	9.59	36.90	231.85	288,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	201.67	1,956,360.00 *gal*
Application event totals		24.88	9.59	36.90	433.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	171.95	1,668,045.00 *gal*
Application event totals		0.00	0.00	0.00	171.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	145.09	1,407,432.00 *gal*
Application event totals		0.00	0.00	0.00	145.09	

DC #2 - 10/27/2022: Oats, silage-soft dough

Field name: DC #2

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

DC #2 - 10/27/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.49	9.47	69.10	749.89	552,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	300.47	7,372,565.00 *gal*
Application event totals		58.49	9.47	69.10	1,050.35	

01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.71	9.74	139.36	578.02	408,000.00 *gal*
Application event totals		43.71	9.74	139.36	578.02	

02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	47.45	10.65	147.56	645.55	432,000.00 *gal*
Application event totals		47.45	10.65	147.56	645.55	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.70	13.75	196.74	816.02	576,000.00 *gal*
Application event totals		61.70	13.75	196.74	816.02	

04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	12.41	4.29	39.49	224.69	432,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	217.57	5,338,536.00 *gal*
Application event totals		12.41	4.29	39.49	442.26	

DC #2 - 06/13/2023: Corn, silage

Field name: DC #2

Crop: Corn, silage

Plant date: 06/13/2023

DC #2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.19	12.52	115.17	655.33	1,260,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	315.89	7,750,965.00 *gal*
Application event totals		36.19	12.52	115.17	971.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	272.08	6,675,965.00 *gal*
Application event totals		0.00	0.00	0.00	272.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.94	15.00	57.74	362.83	1,140,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	237.10	5,817,857.00 *gal*
Application event totals		38.94	15.00	57.74	599.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.58	25.26	97.25	611.09	1,920,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	217.28	5,331,484.00 *gal*
Application event totals		65.58	25.26	97.25	828.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.19	18.95	72.94	458.32	1,440,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	208.12	5,106,594.00 *gal*
Application event totals		49.19	18.95	72.94	666.43	

DC #2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.54	8.68	33.43	210.06	660,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	199.38	4,892,213.00 *gal*
Application event totals		22.54	8.68	33.43	409.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.27	17.05	65.65	412.48	1,296,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	195.87	4,806,116.00 *gal*
Application event totals		44.27	17.05	65.65	608.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.19	18.95	72.94	458.32	1,440,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	176.31	4,326,113.00 *gal*
Application event totals		49.19	18.95	72.94	634.62	

DC #4 - 10/26/2022: Oats, silage-soft dough

Field name: DC #4

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.69	9.99	72.87	790.85	1,056,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	306.89	13,659,516.00 *gal*
Application event totals		61.69	9.99	72.87	1,097.74	

DC #4 - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.67	7.72	56.31	611.11	816,000.00 *gal*
Application event totals		47.67	7.72	56.31	611.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.69	8.84	126.54	524.84	672,000.00 *gal*
Application event totals		39.69	8.84	126.54	524.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.27	8.53	122.02	506.09	648,000.00 *gal*
Application event totals		38.27	8.53	122.02	506.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	20.90	7.47	68.35	440.62	1,320,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	230.02	10,237,890.00 *gal*
Application event totals		20.90	7.47	68.35	670.63	

DC #4 - 06/12/2023: Corn, silage

Field name: DC #4

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

DC #4 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.08	15.25	140.29	798.24	2,784,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	229.75	10,226,148.00 *gal*
Application event totals		44.08	15.25	140.29	1,028.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	221.62	9,863,956.00 *gal*
Application event totals		0.00	0.00	0.00	221.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.24	10.88	41.89	263.19	1,500,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	232.87	10,365,030.00 *gal*
Application event totals		28.24	10.88	41.89	496.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	43.85	13.96	54.44	338.94	1,980,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	216.93	9,655,542.00 *gal*
Application event totals		43.85	13.96	54.44	555.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	53.15	16.92	65.99	410.83	2,400,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	191.12	8,506,544.00 *gal*
Application event totals		53.15	16.92	65.99	601.95	

DC #4 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	180.33	8,026,316.00 *gal*
Application event totals		0.00	0.00	0.00	180.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.15	32.03	123.31	774.83	4,416,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	152.04	6,767,214.00 *gal*
Application event totals		83.15	32.03	123.31	926.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	51.03	16.24	63.35	394.40	2,304,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	174.94	7,786,216.00 *gal*
Application event totals		51.03	16.24	63.35	569.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	162.01	7,210,721.00 *gal*
Application event totals		0.00	0.00	0.00	162.01	

Freeway - 10/27/2022: Oats, silage-soft dough

Field name: Freeway

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Freeway - 10/27/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	189.60	17.70	49.30	939.15	480.00 *ton*
Application event totals		189.60	17.70	49.30	939.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	301.78	5,166,060.00 *gal*
Application event totals		0.00	0.00	0.00	301.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	230.67	3,948,780.00 *gal*
Application event totals		0.00	0.00	0.00	230.67	

Freeway - 06/15/2023: Sudangrass, silage

Field name: Freeway

Crop: Sudangrass, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	161.65	24.71	35.01	902.96	780.00 *ton*
Application event totals		161.65	24.71	35.01	902.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	288.96	4,946,760.00 *gal*
Application event totals		0.00	0.00	0.00	288.96	

Freeway - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	32.80	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	262.23	4,489,050.00 *gal*
Application event totals		32.80	0.00	0.00	262.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	32.80	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	233.60	3,998,970.00 *gal*
Application event totals		32.80	0.00	0.00	233.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	236.36	4,046,211.00 *gal*
Application event totals		0.00	0.00	0.00	236.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	223.04	3,818,261.00 *gal*
Application event totals		0.00	0.00	0.00	223.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	212.32	3,634,621.00 *gal*
Application event totals		0.00	0.00	0.00	212.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	187.07	3,202,403.00 *gal*
Application event totals		0.00	0.00	0.00	187.07	

Freeway - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	159.13	2,724,113.00 *gal*
Application event totals		0.00	0.00	0.00	159.13	

Fullumer - 01/01/2023: Alfalfa Haylage

Field name: Fullumer

Crop: Alfalfa Haylage

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported Solid Manure	Corral solids	256.67	1.17	167.60	5,285.72	200.00 *ton*
Application event totals		256.67	1.17	167.60	5,285.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	272.99	3,115,500.00 *gal*
Application event totals		0.00	0.00	0.00	272.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	243.58	2,779,920.00 *gal*
Application event totals		0.00	0.00	0.00	243.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	228.80	2,611,180.00 *gal*
Application event totals		0.00	0.00	0.00	228.80	

Fullumer - 01/01/2023: Alfalfa Haylage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	225.64	2,575,100.00 *gal*
Application event totals		0.00	0.00	0.00	225.64	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	207.67	2,370,000.00 *gal*
Application event totals		0.00	0.00	0.00	207.67	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	202.39	2,309,760.00 *gal*
Application event totals		0.00	0.00	0.00	202.39	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	189.44	2,162,041.00 *gal*
Application event totals		0.00	0.00	0.00	189.44	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	171.84	1,961,130.00 *gal*
Application event totals		0.00	0.00	0.00	171.84	

09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	151.07	1,724,103.00 *gal*
Application event totals		0.00	0.00	0.00	151.07	

Greenway - 10/27/2022: Oats, silage-soft dough

Greenway - 10/27/2022: Oats, silage-soft dough

Field name: Greenway

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	189.60	17.70	49.30	939.15	688.00 *ton*
Application event totals		189.60	17.70	49.30	939.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	289.74	7,109,276.00 *gal*
Application event totals		0.00	0.00	0.00	289.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	258.53	6,343,489.00 *gal*
Application event totals		0.00	0.00	0.00	258.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	229.13	5,622,207.00 *gal*
Application event totals		0.00	0.00	0.00	229.13	

Greenway - 06/16/2023: Corn, silage

Field name: Greenway

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Greenway - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	161.65	24.71	35.01	902.96	1,118.00 *ton*
Application event totals		161.65	24.71	35.01	902.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	289.74	7,109,276.00 *gal*
Application event totals		0.00	0.00	0.00	289.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	32.80	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	258.53	6,343,489.00 *gal*
Application event totals		32.80	0.00	0.00	258.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	32.80	0.00	0.00	0.00	
T.I.D Canal	Surface water	0.00	0.00	0.00	229.13	5,622,207.00 *gal*
Application event totals		32.80	0.00	0.00	229.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	188.82	4,633,062.00 *gal*
Application event totals		0.00	0.00	0.00	188.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	203.26	4,987,527.00 *gal*
Application event totals		0.00	0.00	0.00	203.26	

Greenway - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	160.44	3,936,771.00 *gal*
Application event totals		0.00	0.00	0.00	160.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	173.07	4,246,721.00 *gal*
Application event totals		0.00	0.00	0.00	173.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	183.40	4,500,100.00 *gal*
Application event totals		0.00	0.00	0.00	183.40	

Lavender - 01/01/2023: Alfalfa Haylage

Field name: Lavender

Crop: Alfalfa Haylage

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	189.60	17.70	49.30	939.15	304.00 *ton*
Application event totals		189.60	17.70	49.30	939.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	259.81	2,816,845.00 *gal*
Application event totals		0.00	0.00	0.00	259.81	

Lavender - 01/01/2023: Alfalfa Haylage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	231.10	2,505,625.00 *gal*
Application event totals		0.00	0.00	0.00	231.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	222.19	2,408,972.00 *gal*
Application event totals		0.00	0.00	0.00	222.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	213.89	2,318,950.00 *gal*
Application event totals		0.00	0.00	0.00	213.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	208.19	2,257,181.00 *gal*
Application event totals		0.00	0.00	0.00	208.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	211.85	2,296,872.00 *gal*
Application event totals		0.00	0.00	0.00	211.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	168.30	1,824,693.00 *gal*
Application event totals		0.00	0.00	0.00	168.30	

Lavender - 01/01/2023: Alfalfa Haylage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	184.52	2,000,513.00 *gal*
Application event totals		0.00	0.00	0.00	184.52	

09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	154.01	1,669,721.00 *gal*
Application event totals		0.00	0.00	0.00	154.01	

Tucker - 01/01/2023: Alfalfa Haylage

Field name: Tucker

Crop: Alfalfa Haylage

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported Solid Manure	Corral solids	256.67	1.17	167.60	5,285.72	270.00 *ton*
Application event totals		256.67	1.17	167.60	5,285.72	

04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	263.28	4,056,291.00 *gal*
Application event totals		0.00	0.00	0.00	263.28	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	244.58	3,768,282.00 *gal*
Application event totals		0.00	0.00	0.00	244.58	

Tucker - 01/01/2023: Alfalfa Haylage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	225.64	3,476,385.00 *gal*
Application event totals		0.00	0.00	0.00	225.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	217.30	3,347,946.00 *gal*
Application event totals		0.00	0.00	0.00	217.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	204.79	3,155,166.00 *gal*
Application event totals		0.00	0.00	0.00	204.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	209.92	3,234,195.00 *gal*
Application event totals		0.00	0.00	0.00	209.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	194.84	3,001,863.00 *gal*
Application event totals		0.00	0.00	0.00	194.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	187.23	2,884,613.00 *gal*
Application event totals		0.00	0.00	0.00	187.23	

Tucker - 01/01/2023: Alfalfa Haylage

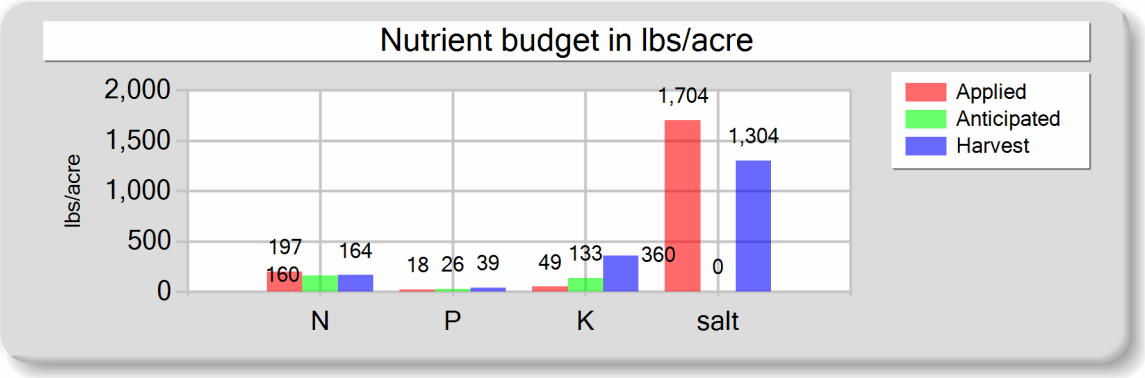
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	170.13	2,621,160.00 *gal*
Application event totals		0.00	0.00	0.00	170.13	

B. NUTRIENT BUDGET

Airport - 10/27/2022: Oats, silage-soft dough

Field name: Airport | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	189.60	17.70	49.30	939.15
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	764.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.60	17.70	49.30	1,703.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	163.53	39.25	359.76	1,303.53
Nutrient balance	33.07	-21.55	-310.47	400.14
Applied to removed ratio	1.20	0.45	0.14	1.31

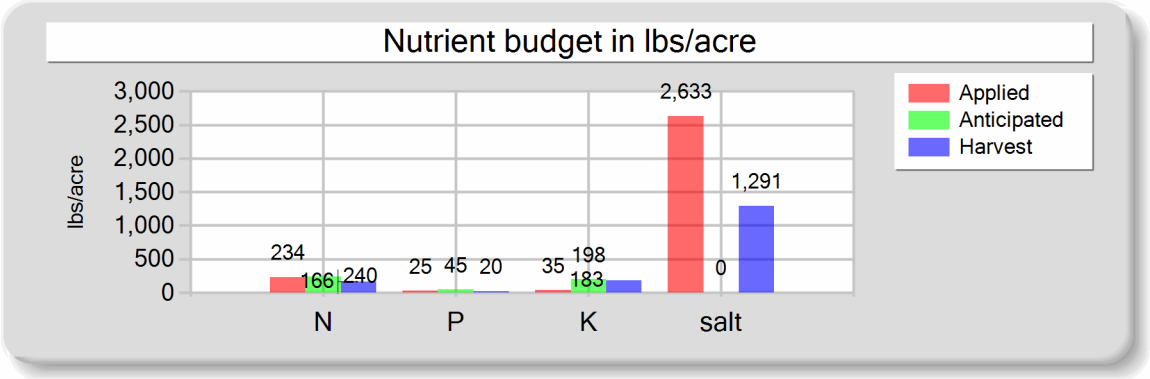
Fresh water applied	
7,416,352.00	*gallons*
273.12	*acre-inches*
16.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Airport - 06/08/2023: Corn, silage

Field name: Airport | Crop: Corn, silage | Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	65.60	0.00	0.00	0.00
Dry manure	161.65	24.71	35.01	902.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,729.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.25	24.71	35.01	2,632.52
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	166.02	20.33	182.96	1,290.92
Nutrient balance	68.22	4.38	-147.96	1,341.60
Applied to removed ratio	1.41	1.22	0.19	2.04

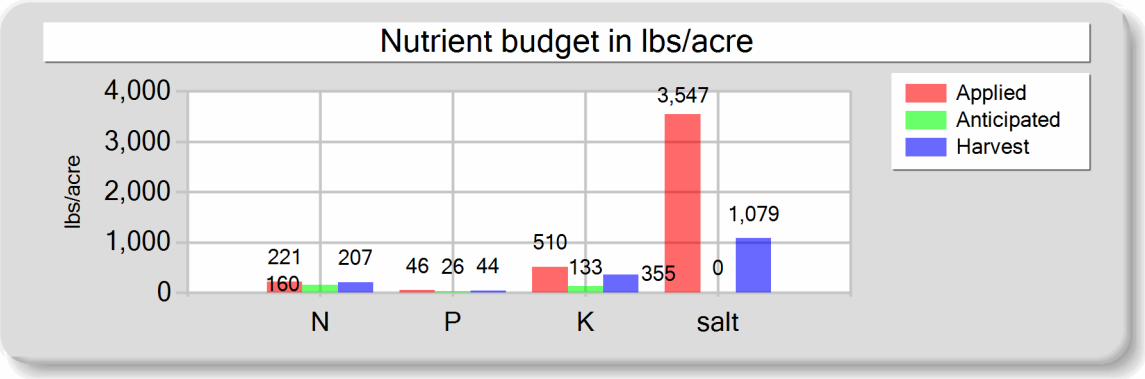
Fresh water applied	
16,777,919.00	*gallons*
617.87	*acre-inches*
36.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC #1 - 10/26/2022: Oats, silage-soft dough

Field name: DC #1 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.79	45.87	509.51	2,997.72
Fresh water	0.00	0.00	0.00	549.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.79	45.87	509.51	3,547.07
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	207.39	43.87	354.95	1,078.98
Nutrient balance	13.40	2.00	154.55	2,468.09
Applied to removed ratio	1.06	1.05	1.44	3.29

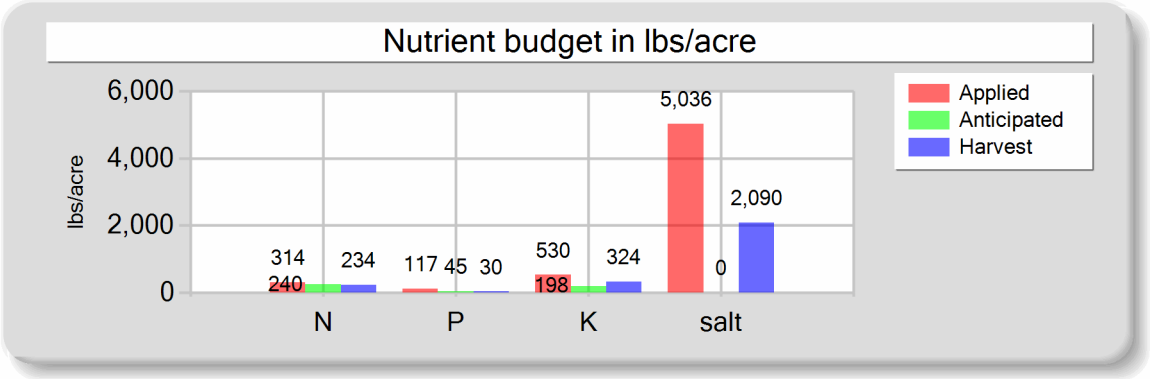
Fresh water applied	
5,329,058.00	*gallons*
196.25	*acre-inches*
11.54	*inches/acre*

Process wastewater applied	
1,056,000.00	*gallons*
38.89	*acre-inches*
2.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC #1 - 06/15/2023: Corn, silage

Field name: DC #1 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	307.18	116.62	529.62	3,245.54
Fresh water	0.00	0.00	0.00	1,790.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.18	116.62	529.62	5,035.78
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	233.68	29.96	323.56	2,090.45
Nutrient balance	80.50	86.66	206.06	2,945.33
Applied to removed ratio	1.34	3.89	1.64	2.41

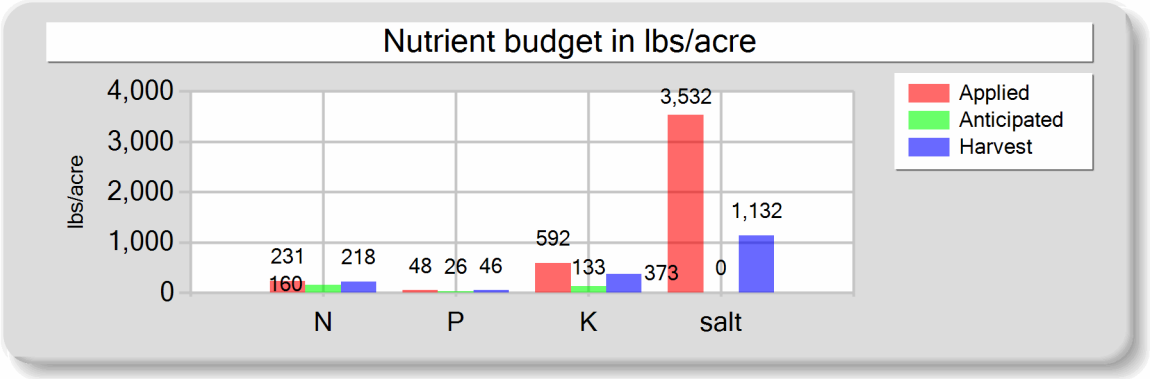
Fresh water applied	
17,366,625.00	*gallons*
639.55	*acre-inches*
37.62	*inches/acre*

Process wastewater applied	
3,651,000.00	*gallons*
134.45	*acre-inches*
7.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC #2 - 10/27/2022: Oats, silage-soft dough

Field name: DC #2 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.76	47.90	592.24	3,014.17
Fresh water	0.00	0.00	0.00	518.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.76	47.90	592.24	3,532.20
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	217.67	46.05	372.56	1,132.49
Nutrient balance	13.09	1.85	219.68	2,399.71
Applied to removed ratio	1.06	1.04	1.59	3.12

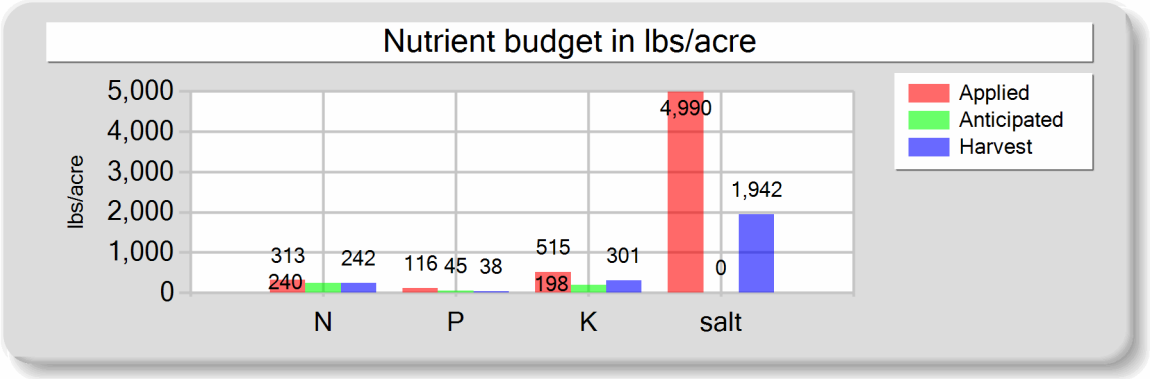
Fresh water applied	
12,711,101.00	*gallons*
468.11	*acre-inches*
10.89	*inches/acre*

Process wastewater applied	
2,400,000.00	*gallons*
88.38	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC #2 - 06/13/2023: Corn, silage

Field name: DC #2 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	305.89	116.41	515.12	3,168.43
Fresh water	0.00	0.00	0.00	1,822.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.89	116.41	515.12	4,990.46
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	241.85	37.62	300.97	1,941.65
Nutrient balance	71.04	78.79	214.15	3,048.81
Applied to removed ratio	1.29	3.09	1.71	2.57

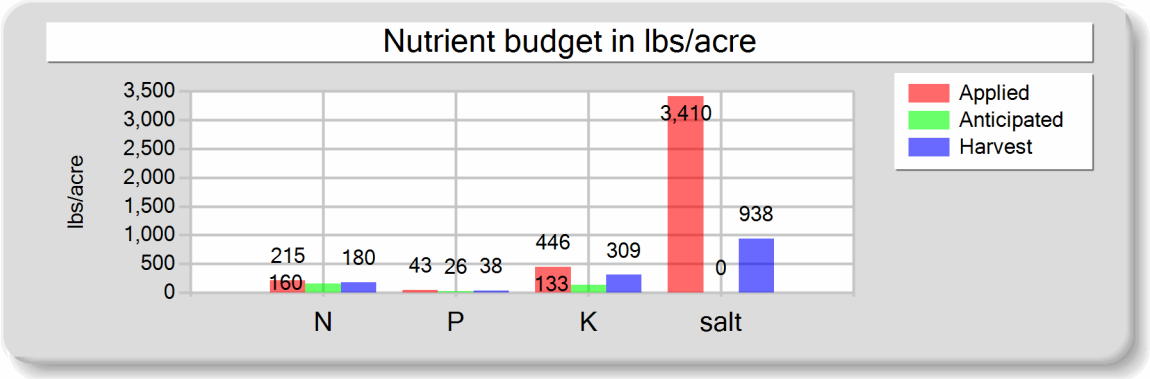
Fresh water applied	
44,707,307.00	*gallons*
1,646.42	*acre-inches*
38.29	*inches/acre*

Process wastewater applied	
9,156,000.00	*gallons*
337.18	*acre-inches*
7.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC #4 - 10/26/2022: Oats, silage-soft dough

Field name: DC #4 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	208.21	42.55	446.09	2,873.50
Fresh water	0.00	0.00	0.00	536.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.21	42.55	446.09	3,410.41
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	180.27	38.13	308.53	937.87
Nutrient balance	34.94	4.41	137.55	2,472.54
Applied to removed ratio	1.19	1.12	1.45	3.64

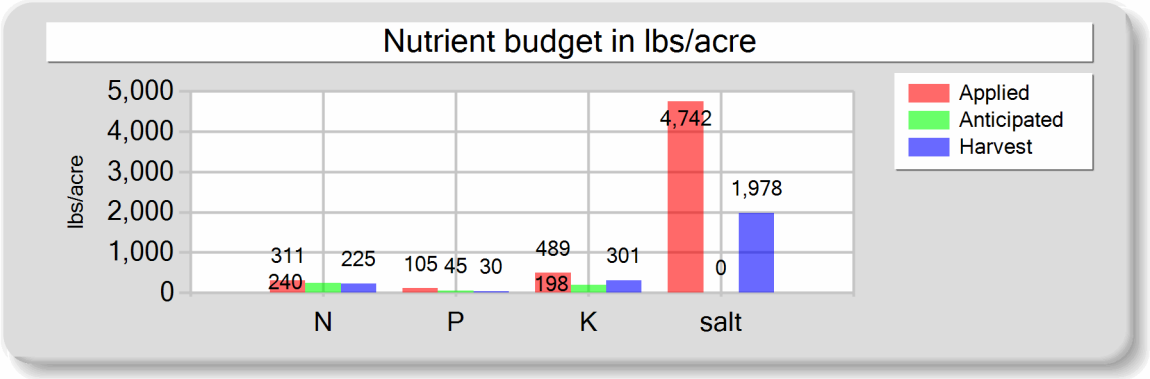
Fresh water applied	
23,897,406.00	*gallons*
880.06	*acre-inches*
11.28	*inches/acre*

Process wastewater applied	
4,512,000.00	*gallons*
166.16	*acre-inches*
2.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC #4 - 06/12/2023: Corn, silage

Field name: DC #4 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	303.51	105.29	489.27	2,980.42
Fresh water	0.00	0.00	0.00	1,761.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.51	105.29	489.27	4,742.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.48	30.06	300.65	1,977.65
Nutrient balance	85.02	75.22	188.62	2,764.38
Applied to removed ratio	1.38	3.50	1.63	2.40

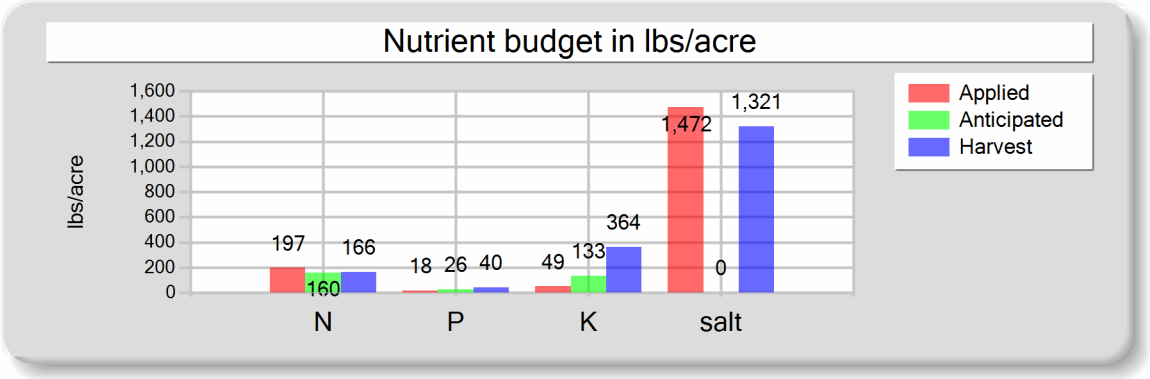
Fresh water applied	
78,407,687.00	*gallons*
2,887.49	*acre-inches*
37.02	*inches/acre*

Process wastewater applied	
15,384,000.00	*gallons*
566.54	*acre-inches*
7.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Freeway - 10/27/2022: Oats, silage-soft dough

Field name: Freeway | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	189.60	17.70	49.30	939.15
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	532.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.60	17.70	49.30	1,471.60
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	165.67	39.76	364.47	1,320.56
Nutrient balance	30.93	-22.06	-315.17	151.03
Applied to removed ratio	1.19	0.45	0.14	1.11

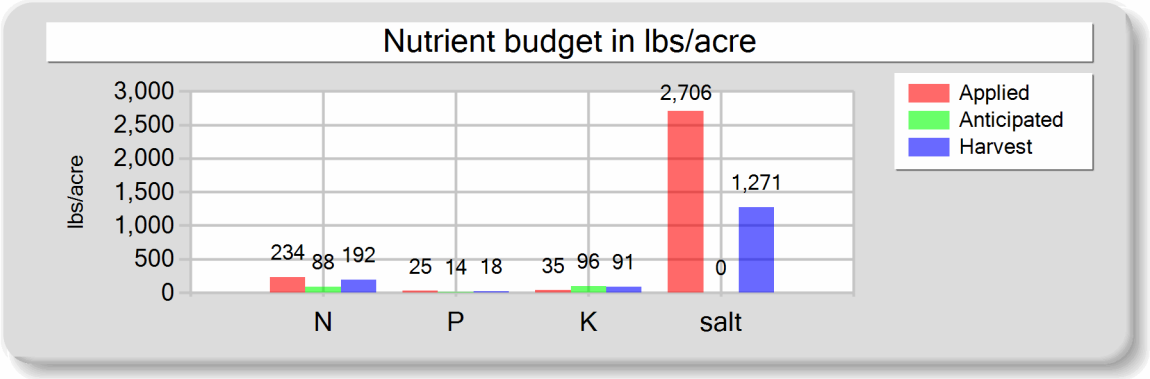
Fresh water applied	
9,114,840.00	*gallons*
335.67	*acre-inches*
11.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Freeway - 06/15/2023: Sudangrass, silage

Field name: Freeway | Crop: Sudangrass, silage | Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	65.60	0.00	0.00	0.00
Dry manure	161.65	24.71	35.01	902.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,802.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.25	24.71	35.01	2,705.67
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	191.62	17.96	91.32	1,271.40
Nutrient balance	42.63	6.75	-56.31	1,434.27
Applied to removed ratio	1.22	1.38	0.38	2.13

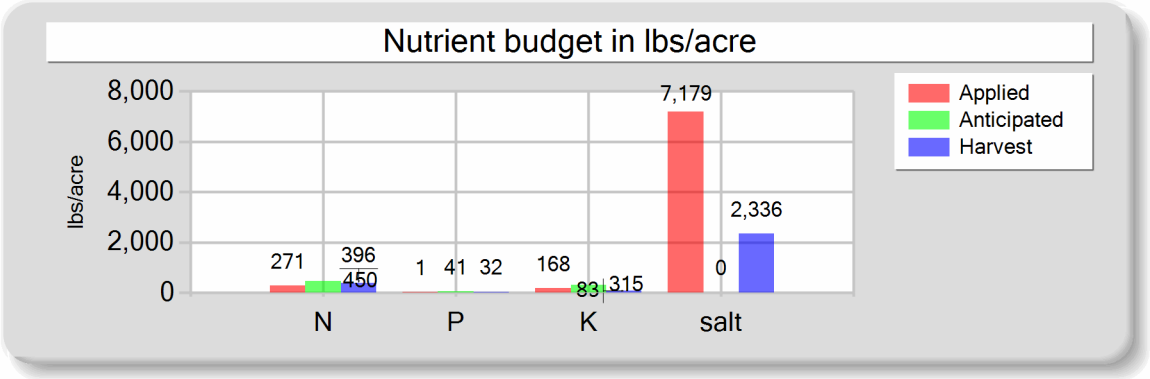
Fresh water applied	
30,860,389.00	*gallons*
1,136.48	*acre-inches*
37.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fullumer - 01/01/2023: Alfalfa Haylage

Field name: Fullumer Crop: Alfalfa Haylage Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	256.67	1.17	167.60	5,285.72
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,893.41
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	270.67	1.17	167.60	7,179.13
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	395.56	32.29	83.42	2,336.40
Nutrient balance	-124.89	-31.12	84.18	4,842.73
Applied to removed ratio	0.68	0.04	2.01	3.07

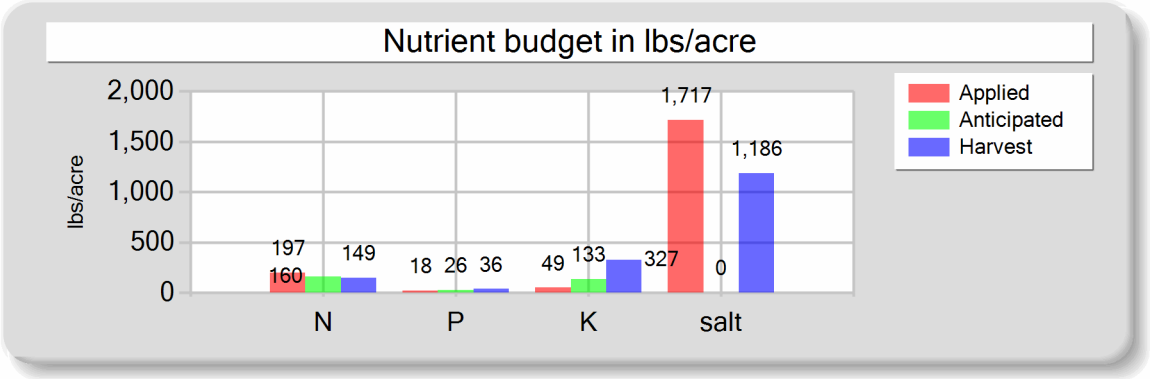
Fresh water applied	
21,608,734.00	*gallons*
795.78	*acre-inches*
39.79	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Greenway - 10/27/2022: Oats, silage-soft dough

Field name: Greenway | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	189.60	17.70	49.30	939.15
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	777.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.60	17.70	49.30	1,716.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	148.84	35.72	327.44	1,186.41
Nutrient balance	47.76	-18.02	-278.15	530.13
Applied to removed ratio	1.32	0.50	0.15	1.45

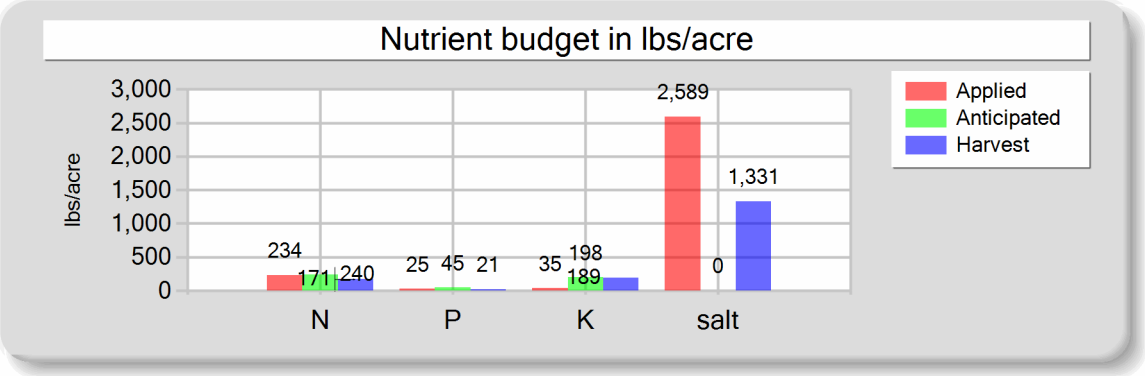
Fresh water applied	
19,074,972.00	*gallons*
702.47	*acre-inches*
16.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Greenway - 06/16/2023: Corn, silage

Field name: Greenway Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	65.60	0.00	0.00	0.00
Dry manure	161.65	24.71	35.01	902.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,686.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.25	24.71	35.01	2,589.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	171.21	20.96	188.68	1,331.21
Nutrient balance	63.04	3.75	-153.67	1,258.14
Applied to removed ratio	1.37	1.18	0.19	1.95

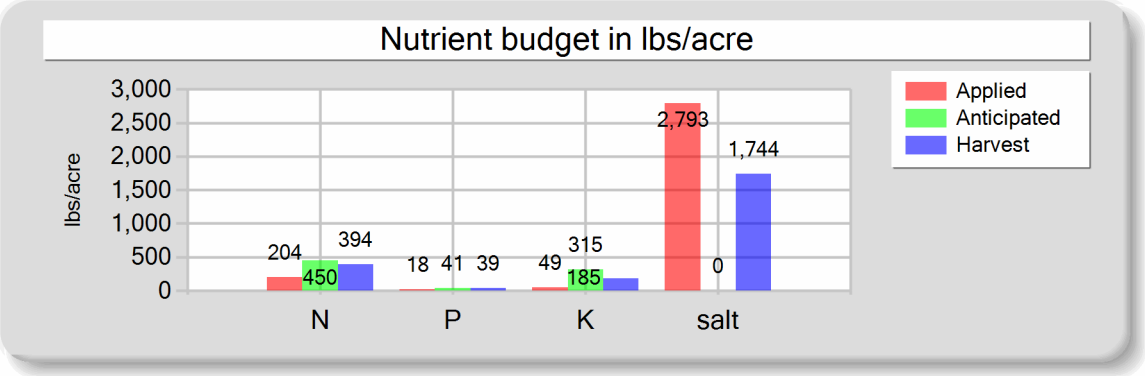
Fresh water applied	
41,379,153.00	*gallons*
1,523.85	*acre-inches*
35.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lavender - 01/01/2023: Alfalfa Haylage

Field name: Lavender | Crop: Alfalfa Haylage | Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	189.60	17.70	49.30	939.15
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,853.85
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	203.60	17.70	49.30	2,793.00
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	394.04	38.63	185.43	1,743.98
Nutrient balance	-190.44	-20.94	-136.14	1,049.02
Applied to removed ratio	0.52	0.46	0.27	1.60

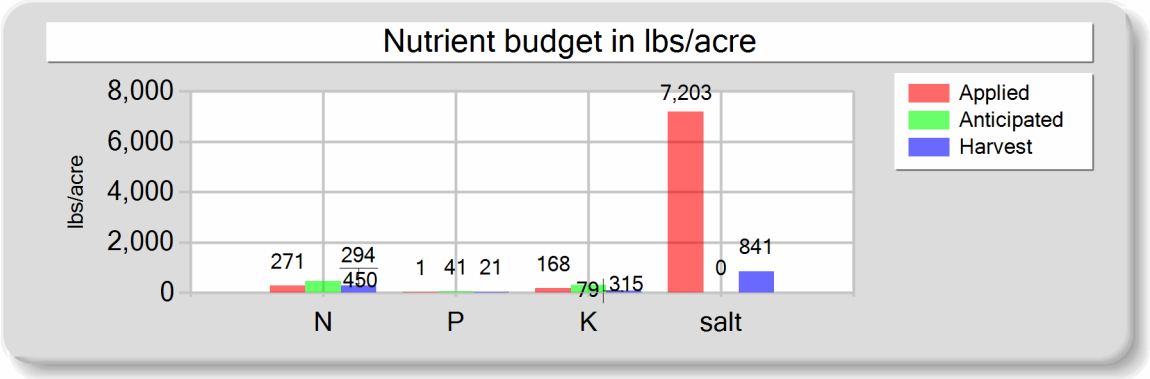
Fresh water applied	
20,099,372.00	*gallons*
740.19	*acre-inches*
38.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tucker - 01/01/2023: Alfalfa Haylage

Field name: Tucker | Crop: Alfalfa Haylage | Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	256.67	1.17	167.60	5,285.72
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,917.69
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	270.67	1.17	167.60	7,203.41
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	294.13	20.71	78.71	840.69
Nutrient balance	-23.46	-19.54	88.89	6,362.72
Applied to removed ratio	0.92	0.06	2.13	8.57

Fresh water applied	
29,545,901.00	*gallons*
1,088.08	*acre-inches*
40.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Corral Solids

Sample and source description: 2022 Corral Solids

Sample date: 11/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	1,400.00	3,900.00	7,200.00	1,600.00	300.00	2,100.00	433.00		7.43
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Fall 2022 Separated Solids

Sample and source description: Fall 2022 Separated Solids

Sample date: 11/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,300.00	6,700.00	7,800.00							23.90
DL	100.00	100.00	100.00							0.01

Imported 2022 Solid Manure

Sample and source description: Imported 2022 Solid Manure

Sample date: 11/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	100.00	14,300.00							45.10
DL	100.00	100.00	100.00							0.01

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,100.00	1,900.00	2,200.00							15.20
DL	100.00	100.00	100.00							0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	2,400.00	3,400.00							8.77
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 21.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	3,800.00	17,100.00							45.00
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	2,200.00	3,800.00	7,800.00	3,000.00	800.00	2,100.00	496.00		6.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	546.00	302.00	0.00	0.00	88.40	645.00								4,670.00	7,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ1

Sample and source description: Imported WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	566.00	286.00	0.00	0.00	127.00	1,760.00								8,270.00	7,700
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	552.00	274.00	0.00	0.00	123.00	1,760.00								8,280.00	7,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ2

Sample and source description: Imported WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	148.00	104.00	0.00	0.00	52.90	484.00								2,850.00	3,120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	148.00	104.00	0.00	0.00	51.20	471.00								2,820.00	2,680
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ3

Sample and source description: Imported WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	207.00	120.00	0.00	0.00	65.90	257.00	77.90	56.20	62.30	1,100.00	0.00	83.70	117.00	2,610.00	1,600
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	176.00	123.00	0.00	0.00	67.80	261.00	80.00	57.00	63.70	1,080.00	0.00	79.70	118.00	2,610.00	1,640
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	792.00	510.00	0.00	0.00	75.70	1,170.00								9,760.00	10,200
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Irrigation Well

Chris Ag Well

Sample description: Chris Ag Well

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	24.90	0.00	19.30								774.00	450
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

South Deep Well

Sample description: South Deep Well

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	44.50	0.00	41.70								1,060.00	660
DL	0.50	0.50	0.50								1.00	10

T.I.D Canal

Lateral 5

Sample description: Lateral 5

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	210
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Airport - 10/27/2022: Oats, silage-soft dough

Airport - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,100.00	1,000.00	6,900.00		5.20
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	1,200.00	11,000.00		9.40
DL	100.00	100.00	10.00		0.01

Airport - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 53.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	1,200.00	3,700.00		8.80
DL	100.00	100.00	100.00		0.01

DC #1 - 10/26/2022: Oats, silage-soft dough

DC #1 - 10/26/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	1,900.00	17,600.00		9.30
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,900.00	1,300.00	13,000.00		11.30
DL	100.00	100.00	100.00		0.01

DC #1 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	500.00	5,400.00		9.80
DL	100.00	100.00	100.00		0.01

DC #2 - 10/27/2022: Oats, silage-soft dough

DC #2 - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,100.00	8,900.00		8.10
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,000.00	10,100.00		10.10
DL	100.00	100.00	100.00		0.01

DC #2 - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	700.00	5,600.00		9.10
DL	100.00	100.00	100.00		0.01

DC #2 - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	600.00	6,000.00		9.20
DL	100.00	100.00	100.00		0.01

DC #4 - 10/26/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,100.00	1,400.00	14,800.00		9.60
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,900.00	1,300.00	13,000.00		11.30
DL	100.00	100.00	100.00		0.01

DC #4 - 06/12/2023: Corn, silage

DC #4 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	400.00	5,400.00		8.70
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	600.00	5,400.00		10.00
DL	100.00	100.00	100.00		0.01

Freeway - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	1,200.00	11,000.00		9.40
DL	100.00	100.00	100.00		0.01

Freeway - 06/15/2023: Sudangrass, silage

Freeway - 06/15/2023: Sudangrass, silage

Sudan Silage

Sample and source description: Sudan Silage

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 43.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,200.00	6,100.00		14.90
DL	100.00	100.00	100.00		0.01

Fullumer - 01/01/2023: Alfalfa Haylage

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 47.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	2,400.00	184,000.00		11.60
DL	100.00	100.00	100.00		0.01

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 08/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 44.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	1,600.00	12,600.00		13.10
DL	100.00	100.00	100.00		0.01

Fullumer - 01/01/2023: Alfalfa Haylage

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 50.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	1,200.00	3,100.00		17.40
DL	100.00	100.00	100.00		0.01

Greenway - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	900.00	7,200.00		5.80
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	1,200.00	11,000.00		9.40
DL	100.00	100.00	100.00		0.01

Greenway - 06/16/2023: Corn, silage

Greenway - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,100.00	500.00	2,200.00		7.80
DL	100.00	100.00	100.00		0.01

Lavender - 01/01/2023: Alfalfa Haylage

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 44.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	2,000.00	21,400.00		12.60
DL	100.00	100.00	100.00		0.01

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 08/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 43.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	2,400.00	17,800.00		6.20
DL	100.00	100.00	100.00		0.01

Lavender - 01/01/2023: Alfalfa Haylage

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	1,000.00	4,800.00		11.40
DL	100.00	100.00	100.00		0.01

Tucker - 01/01/2023: Alfalfa Haylage

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 43.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	2,200.00	24,800.00		11.90
DL	100.00	100.00	100.00		0.01

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 08/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 46.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	2,000.00	8,400.00		10.20
DL	100.00	100.00	100.00		0.01

Tucker - 01/01/2023: Alfalfa Haylage

Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 51.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	1,000.00	3,800.00		8.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

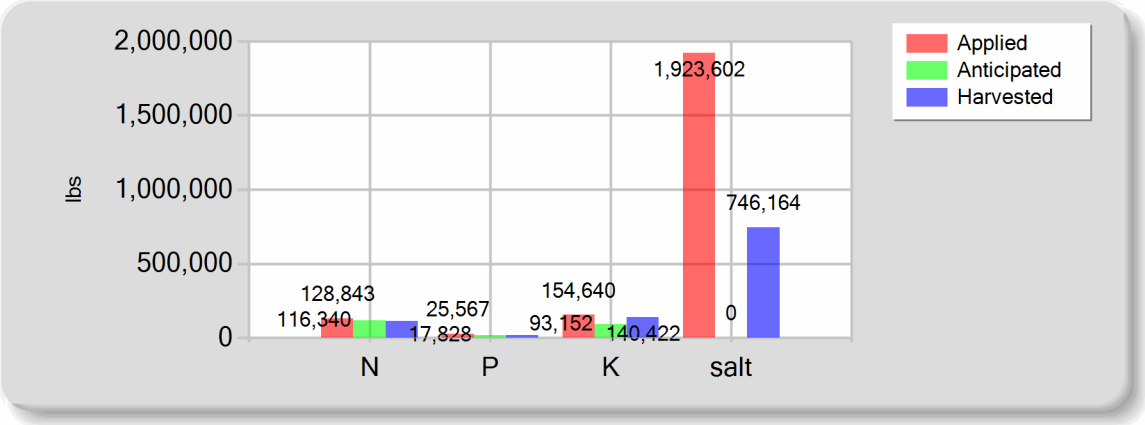
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

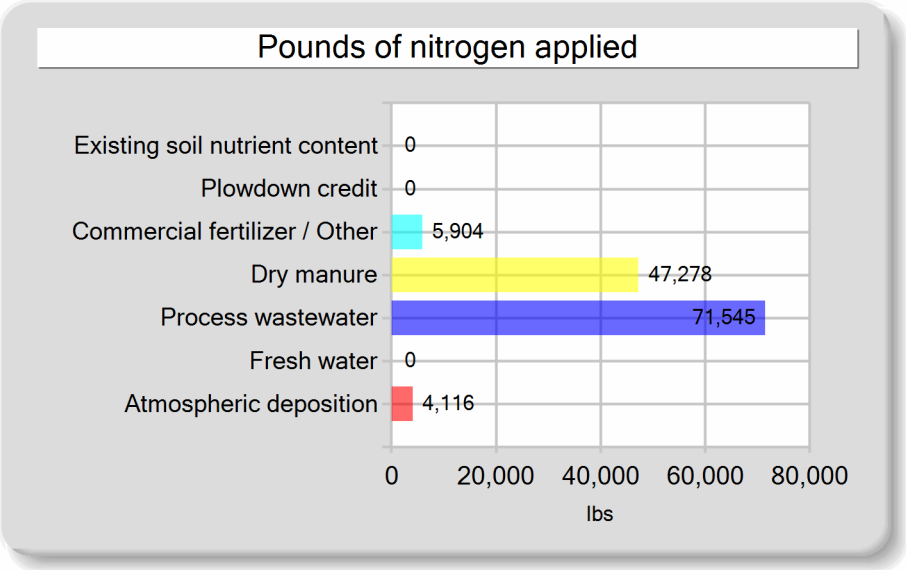
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

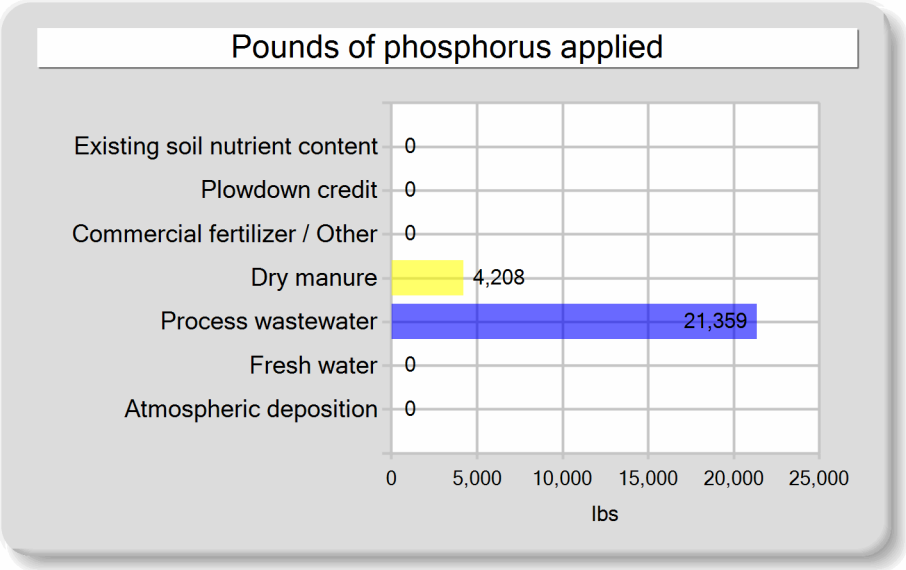
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5,904.00	0.00	0.00	0.00
Dry manure	47,278.04	4,207.88	16,400.85	432,062.74
Process wastewater	71,545.32	21,358.91	138,238.99	828,593.41
Fresh water	0.00	0.00	0.00	662,946.26
Atmospheric deposition	4,116.00	0.00	0.00	0.00
Total nutrients applied	128,843.37	25,566.79	154,639.84	1,923,602.40
Anticipated crop nutrient removal	116,340.00	17,827.80	93,152.40	0.00
Actual crop nutrient removal	112,328.77	17,293.33	140,422.30	746,163.86
Nutrient balance	16,514.59	8,273.46	14,217.55	1,177,438.54
Applied to removed ratio	1.15	1.48	1.10	2.58

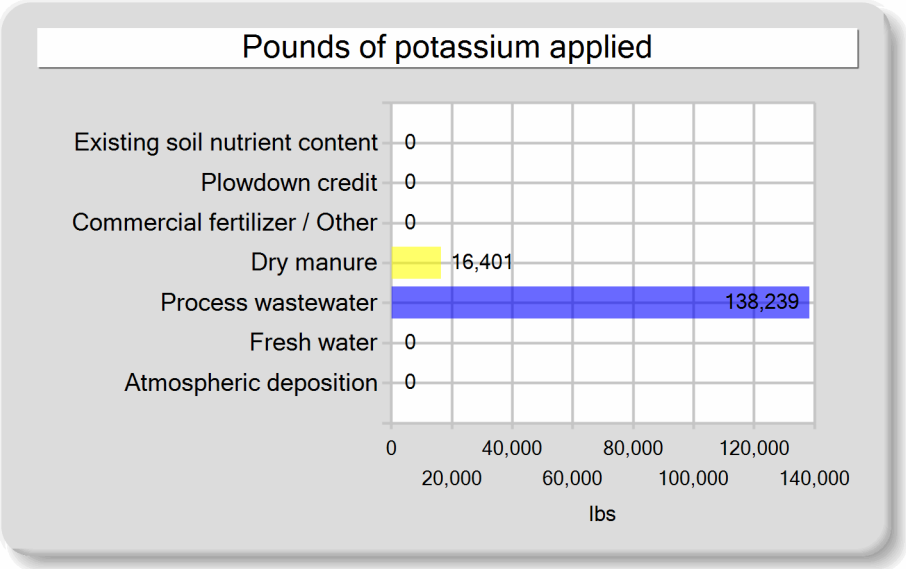
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

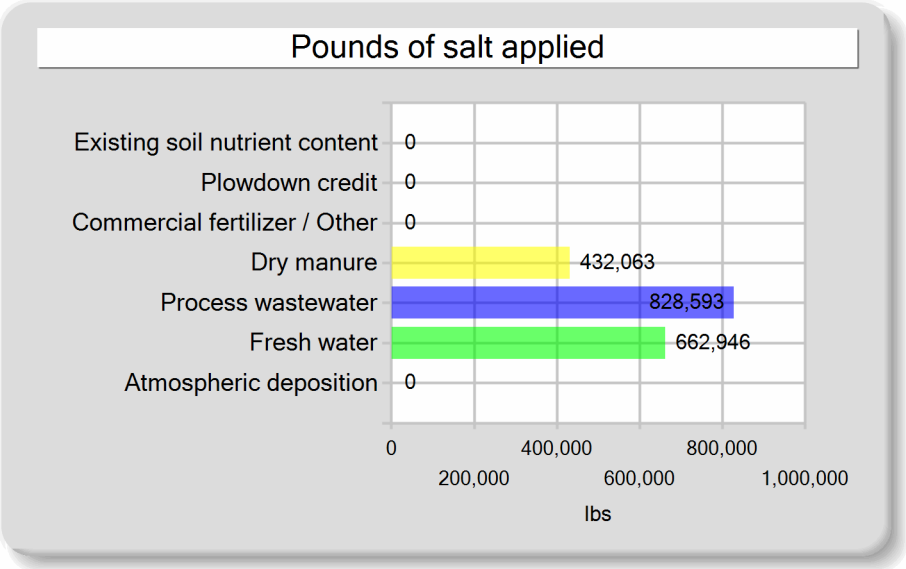


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mark Ahlem Mark Ahlem (May 21, 2024 14:08 PDT)	Mark Ahlem Mark Ahlem (May 21, 2024 14:08 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mark Ahlem	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 21, 2024	May 21, 2024
DATE	DATE

ma

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double Cross #2
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
JLC House	T3291016A	Electrical Conductivity (EC)	EPA 120.1 and	0.316	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.69	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	202	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn	T3291016B	Electrical Conductivity (EC)	EPA 120.1 and	0.313	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.58	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	201	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Chris House	T3291016C	Electrical Conductivity (EC)	EPA 120.1 and	0.767	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	491	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
South Deep Well	T3291016D	Electrical Conductivity (EC)	EPA 120.1 and	1.06	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	41.7	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	660	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	44.5	mg/L
		Soluble Salts (SALT-SOL)	Calculation	678	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)			1.23
Electrical Conductivity (EC)	97.6			0.316 (T3291016A)	0.317 (Dup)	0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)			2.88
Total Dissolved Solids (TDS)	93.5			1190 (T3296001E)	1150 (Dup)	3.43
Total Nitrogen (TN)	98.6			8.40 (T3291009B)	8.40 (null)	0.000

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Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Chris Ag Well	T3291016E	Electrical Conductivity (EC)	EPA 120.1 and	0.774	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.3	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	450	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	24.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	496	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)			1.23
Electrical Conductivity (EC)	97.6			0.316 (T3291016A)	0.317 (Dup)	0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)			2.88
Total Dissolved Solids (TDS)	93.5			1190 (T3296001E)	1150 (Dup)	3.43
Total Nitrogen (TN)	98.6			8.40 (T3291009B)	8.40 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DC#2
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: SB
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	10/10	3:00	Domestic well	JLC House		15.2'	✓				
2		3:15	↓	Milk Barn		14.8'	✓				
3		3:30	↓	Chris House		16.0'	✓				
4		3:45	Irrigation well	South Deep well		15.0'					H7
5	✓	4:00	↓	Chris Ag well		15.6'					H7
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged in By:	LABORATORY NUMBER
1st		FRAG				
2nd		DENELE ANALYTICAL, INC.	10-18-23 9:45			T3291016

Required to be submitted to any agency? (YES / NO)

2023 Annual Report Double Cross #2

Final Audit Report 2024-05-21

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